🛡 SOC ANALYST LAB REPORT

Email Phishing Analysis

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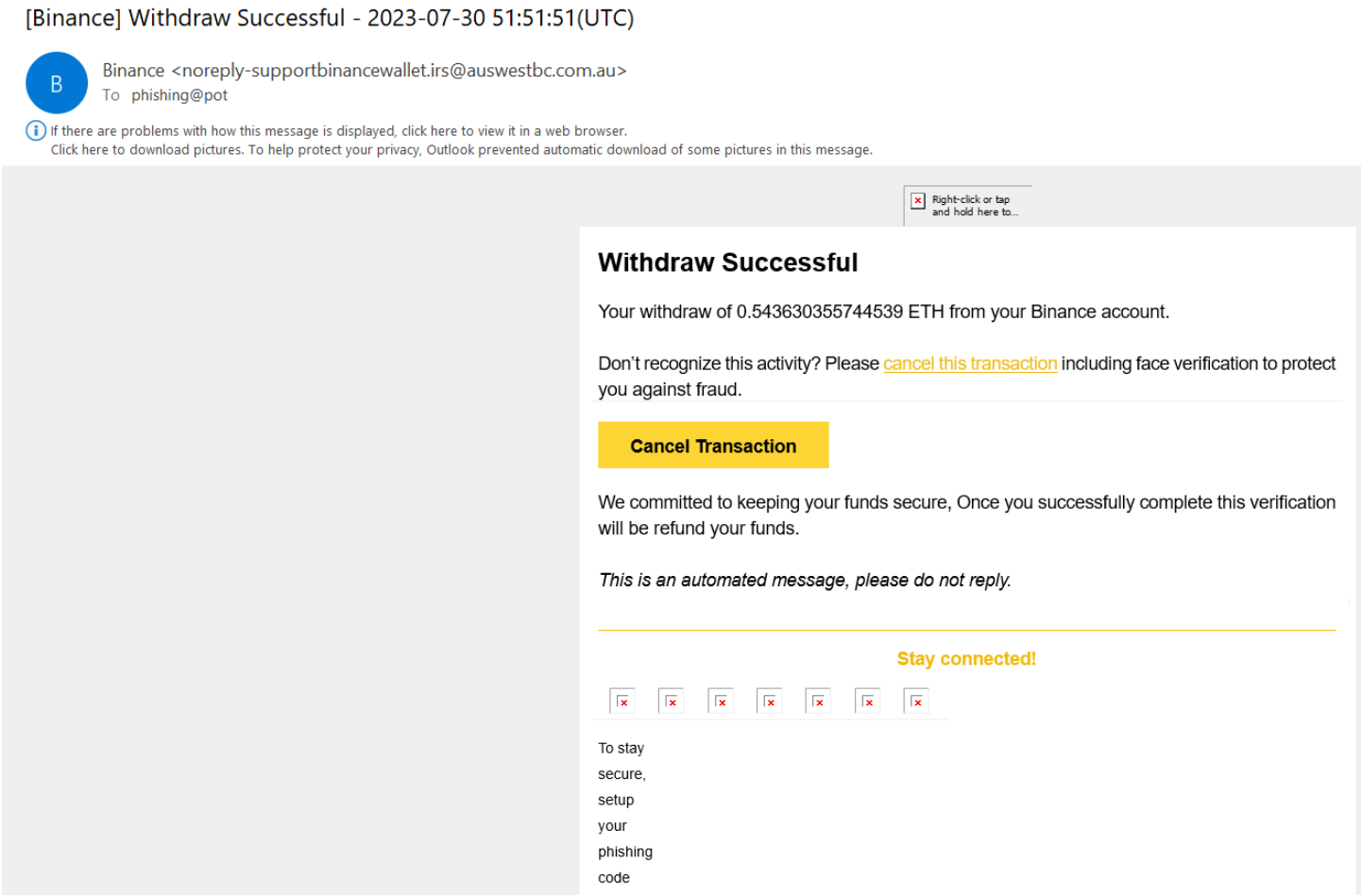
1. Executive Summary

This report documents a Tier-1 SOC investigation into a suspected phishing email designed to steal cryptocurrency credentials. The investigation confirmed that the email is a highly deceptive phishing attempt leveraging Binance brand impersonation. The attacker utilized an Amazon AWS infrastructure IP (69.169.224.12) and successfully bypassed basic SPF and DKIM checks using a completely mismatched Return-Path domain. The email contains a malicious call-to-action link (shylshom.com) designed for credential harvesting. Immediate blocking of the associated IOCs and a retroactive sweep of the email gateway are recommended.



2. Tools & Technologies

The following tools were employed across all investigation scenarios:

- MXToolbox — Analyzing Email Headers
- VirusTotal — IP/domain threat intelligence lookup
- MITRE ATT&CK Framework — Tactic and technique mapping

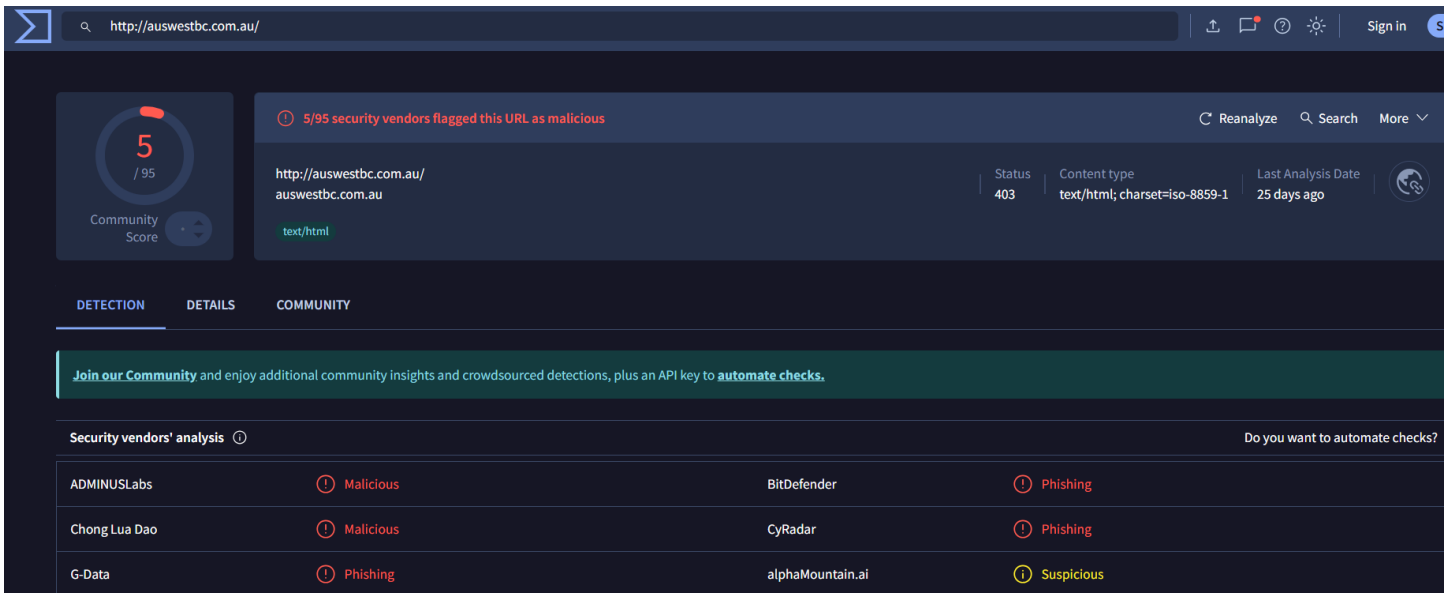
3. Scenario 1 — Phishing Email Analysis

3.1 Email Overview

The From Address of the Email does not match the Binance domain or Binance related email.

Date	Sun, 30 Jul 2023 23:57:35 +0000
To	phishing@pot
From	Binance <noreply-supportbinancewallet.irs@auswestbc.com.au>
Subject	[Binance] Withdraw Successful - 2023-07-30 51:51:51(UTC)

The From Address of the Email domain is also reported Malicious.



3.2 Header Analysis

The originating IP address was identified as **69.169.224.12**, which is associated with Amazon infrastructure and not with Binance

```
OrgName:        Amazon Web Services, Inc.
OrgId:          AMAZO-22
Address:        Amazon Web Services
Address:        1200 12th Avenue South
City:           Seattle
StateProv:      WA
PostalCode:     98144
Country:        US
RegDate:        2010-03-08
Updated:        2026-04-17
Ref:            https://rdap.arin.net/registry/entity/AMAZO-22
```

This email does not have a reply-to enabled.

However, the Return-Path address differs completely from the visible From address, which is a classic indicator of domain spoofing.

3.3 Authentication Analysis

SPF Result: Passed

DKIM Result: Passed

DMARC Result: No policy

Even though SPF and DKIM are passed they are not aligned meaning the email sending server and DKIM signature are not from the actual domain. They are not aligned

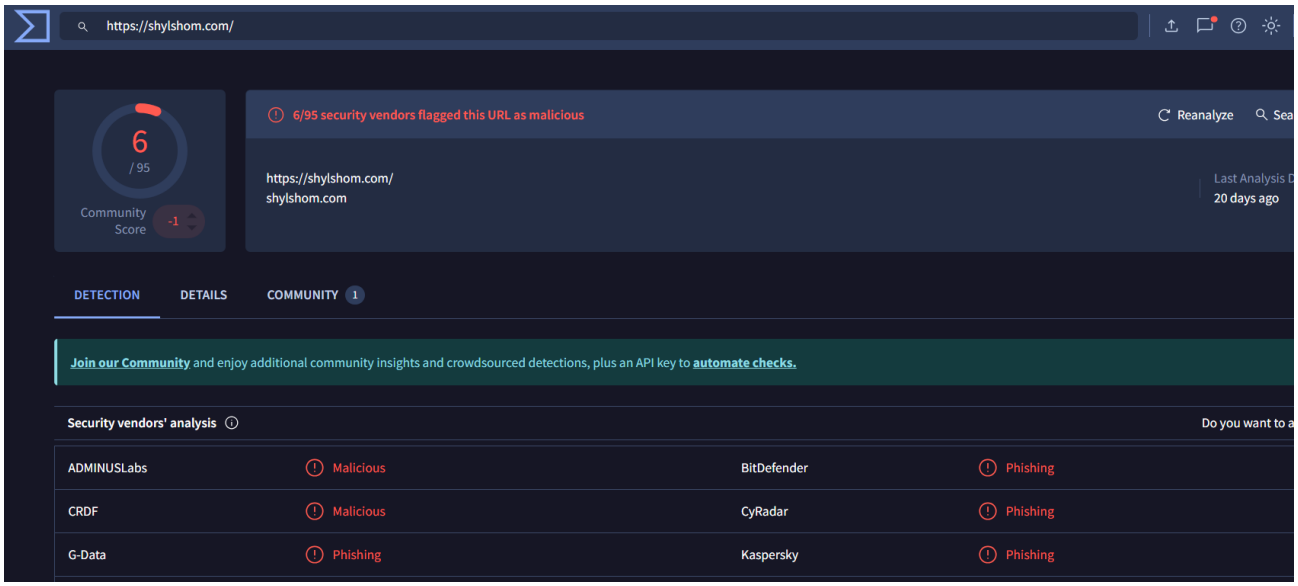
- ✖ DMARC Compliant (No DMARC Record Found)
  - ✖ SPF Alignment
  - ✔ SPF Authenticated
  - ✖ DKIM Alignment
  - ✖ DKIM Authenticated

3.4 URL Analysis

The email contains a URL to cancel a transaction: shylshom.com

This technique is used to deceive users into trusting malicious domains

This URL is also Malicious in VirusTotal Threat Intelligence



3.5 MITRE ATT&CK Mapping

Tactic: Credential Access (TA0006)

Technique: T1566 - Phishing

Sub-technique: T1566.002 - Spear phishing Link

Description: The adversary utilized a deceptive email masquerading as a Binance withdrawal alert. The email contains a malicious link designed to direct the victim to a credential harvesting site under the guise of cancelling a fraudulent transaction.

4. Risk Assessment

Risk Category	Level	Indicator	Priority
Credential Theft	HIGH	Malicious URL (shylshom.com) disguised as a cancellation link.	Immediate
Brand Impersonation	HIGH	Spoofed Binance display name and graphics.	Immediate
Gateway Evasion	MEDIUM	Attacker domain passed SPF/DKIM but failed DMARC alignment.	High

5. Recommendations

1. **Block the malicious domain at email gateway**
2. **Enable DMARC enforcement policies.**
3. **Use URL filtering and sandboxing**.
4. **Monitor email logs for similar patterns**.
5. **User Awareness, Utilize this email template in the next internal phishing simulation to train employees.**

6. Conclusion

The investigation confirms that email is a phishing attempt leveraging domain spoofing and social engineering techniques.

Although no attachment was present, the use of a malicious link indicates a credential harvesting attack.

This incident should be escalated for further monitoring and preventive action.